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Optimizing Sucker Rod Pump Selection for Mature Oilfields: A Case Study on Pump Size and Clearance

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Abstract

In mature oilfields, optimizing sucker rod pump selection is a critical factor for maximizing production efficiency, equipment reliability, and operational run life.

The operator of the field has conducted a comprehensive series of field trials to compare the performance of smaller pump sizes versus larger ones, as well as pumps with varying plunger clearances, under diverse well conditions—including differences in production rates, reservoir characteristics, and fluid properties.

The objective of these trials was to assess pump performance across key metrics such as volumetric efficiency, fluid handling capability, pump fillage, and overall system reliability.

A systematic design approach was adopted, deploying pumps ranging in size from 1.25" to 2.75" and using different plunger clearances from 0.008" to 0.01" to evaluate performance, particularly under gas interference conditions. These trials were conducted in a wide range of challenging well environments. The evaluation was based on an integrated set of criteria, encompassing both surface and downhole parameters, to ensure robust and reliable outcomes.

The results revealed that smaller pumps consistently outperformed larger ones in wells with lower bottom-hole pressure and significant gas interference. These pumps maintained higher pump fillage due to better gas handling capability, resulting in more stable operations and a reduction in pump-off conditions.

Additionally, pumps with higher clearance, although experiencing increased slippage, effectively mitigated gas lock and contributed to improved pump fillage and continuous production. This translated into longer operational durations, reduced maintenance costs, and enhanced equipment reliability.

A noticeable improvement in production was also observed after installing smaller pumps with higher clearance—some wells recorded production increases of up to 140%. This study presents a detailed performance analysis covering efficiency, wear patterns, failure modes, and the economic impact of varying pump sizes and clearances. The findings underscore the importance of selecting the optimal pump size and clearance to enhance the efficiency of beam pump artificial lift systems. The paper offers a systematic design and selection methodology based on real-world data from over 300 wells, providing valuable insights for optimizing artificial lift in depleted, low-rate wells.

Introduction

Background

The operator manages a mature oil field in Oman, characterized by a high Gas-Oil Ratio and declining reservoir pressure. Artificial lift, particularly sucker rod pumps, is critical to sustaining production in this environment. However, the presence of free gas at the pump intake introduces several operational challenges.

Common issues include gas locking, pump-off conditions, and inefficient fluid load-up. These conditions lead to reduced pump fillage, directly impacting gross fluid production. In addition, incomplete fillage increases the likelihood of rod string buckling due to inadequate fluid cushioning and unstable loading conditions.

These challenges are typically more severe when larger displacement rod pumps are used, as they require higher intake volumes and are more sensitive to gas interference.

Objective

This study aims to evaluate the effectiveness of using smaller-capacity sucker rod pumps with high clearance in high GOR oil wells within the operator's field. The objective is to assess their impact on mitigating gas interference, improving pump fillage, enhancing gross production, and reducing the risk of mechanical failures such as rod buckling.

Field Description

Field Overview

The oil field, operated by the field's operator, is located in western Oman within Block 5 and has been in production since the late 1980s. It is a mature, onshore carbonate reservoir primarily producing from the Natih and Shuaiba formations. The field is known for its heterogeneous reservoir quality, complex fracture systems, and naturally driven production mechanisms in its early life, transitioning over time to artificial lift support due to declining reservoir pressure.

Oil production in the field is maintained through a combination of secondary recovery techniques, including waterflooding, and an extensive artificial lift program, predominantly based on sucker rod pumps and ESPs. The field has undergone several phases of development, including horizontal drilling, selective stimulation, and zonal management, to maximize recovery from tight and compartmentalized reservoir intervals.

The operator has consistently implemented data-driven optimization strategies, integrating surveillance, diagnostics, and production analytics to improve field performance. This has made the field a recognized example of cost-effective mature field management in the region, balancing operational efficiency with long-term reservoir sustainability.

Reservoir and Well Context

[Table 1](#) provides an overview of the key reservoir and well parameters relevant to this study, establishing the context for the subsequent analysis of pump performance and operational challenges in the field.

Table 1—Average Field - Well Overview

Parameter	Value (Avg.)
Reservoir Name and Type	Natih / Shuaiba – Carbonate
Reservoir Depth	800 to 1800 m TVD
Type of Wells	Horizontal, Deviated, and Vertical
Artificial Lift Type	Sucker Rod Pump, ESP
Gas-Oil Ratio (GOR)	0 to +4000 scf/STB
Average Gross Production	10 to 1000 bpd
API Gravity	~38 API (Shuiba) & ~28 API (Natih)

Artificial Lift & Completion Description

The Field employs a range of artificial lift systems to sustain production in its mature, high GOR carbonate reservoir environment. Sucker rod pumps (SRP) are the dominant artificial lift method, deployed in the majority of wells due to their reliability, adaptability to various production rates, and cost-effectiveness in onshore operations. Electrical submersible pumps (ESP) are selectively used in higher-rate producers or in wells where rod pump deployment is less feasible.

The downhole pumps in sucker rod completions are mainly of two types: insert pumps and tubing pumps. Insert pumps are preferred when easy retrieval and maintenance without pulling the tubing is required, making them suitable for wells with frequent intervention needs. Tubing pumps are used where a larger pump bore is desired. Before running this project, pump sizes typically used in this field were 1.75", 2.25", and 2.75". The sucker rods are of tapered type, with diameters of 1", 7/8", and 3/4", complemented by a 1.5" sinker bar to stabilize the rod string and improve stroke efficiency. Tubing sizes in use are 2 7/8" and 3.5", selected based on well design, target rates, and completion constraints. The pump clearance previously used is 0.002" & 0.008" which is the space between the barrel and plunger.

Table 2 presents the ideal production capacities and corresponding volumetric efficiencies of various sucker rod pump sizes, assuming a fixed reservoir inflow of 90 bbl/day. The production rates are based on a standard operating setup (197-inch stroke length, 3 SPM, and 100% theoretical fillage). While larger pumps offer higher theoretical output, the calculated volumetric efficiency decreases significantly as pump size increases. This highlights a key operational trade-off: smaller pumps operate closer to their displacement capacity in low-inflow wells, resulting in higher efficiency, whereas larger pumps tend to be underfilled and less efficient under the same inflow conditions. These insights are especially critical for pump selection in viscous, low-GOR environments, where maintaining high fillage is challenging.

Table 2—Ideal Production Capacities and Corresponding Volumetric Efficiencies

Pump Size (inches)	Production Rate at Ideal Conditions (bbl/day)	Volumetric Efficiency (%), assuming a fixed reservoir inflow of 90 bbl/day
1.25	107	84.1%
1.5	155	58.1%
1.75	211	42.7%
2.25	348	25.9%
2.75	521	17.3%

The following list represent the range of sucker rod pump's sizes and types utilized in the field:

1. 25-175-RAHM-24-4-2, 1.75" Insert Pump
2. 30-225-RAHM-24-4-2, 2.25" Insert Pump
3. 25-175-THM-24-4-4, 1.75" Tubing Pump
4. 25-225-THM-24-4-4, 2.25" Tubing Pump
5. 30-275-THM-24-4-4, 2.75" Tubing Pump

Figure 1 is showing a typical schematic of a vertical installation of the beam pump completion.

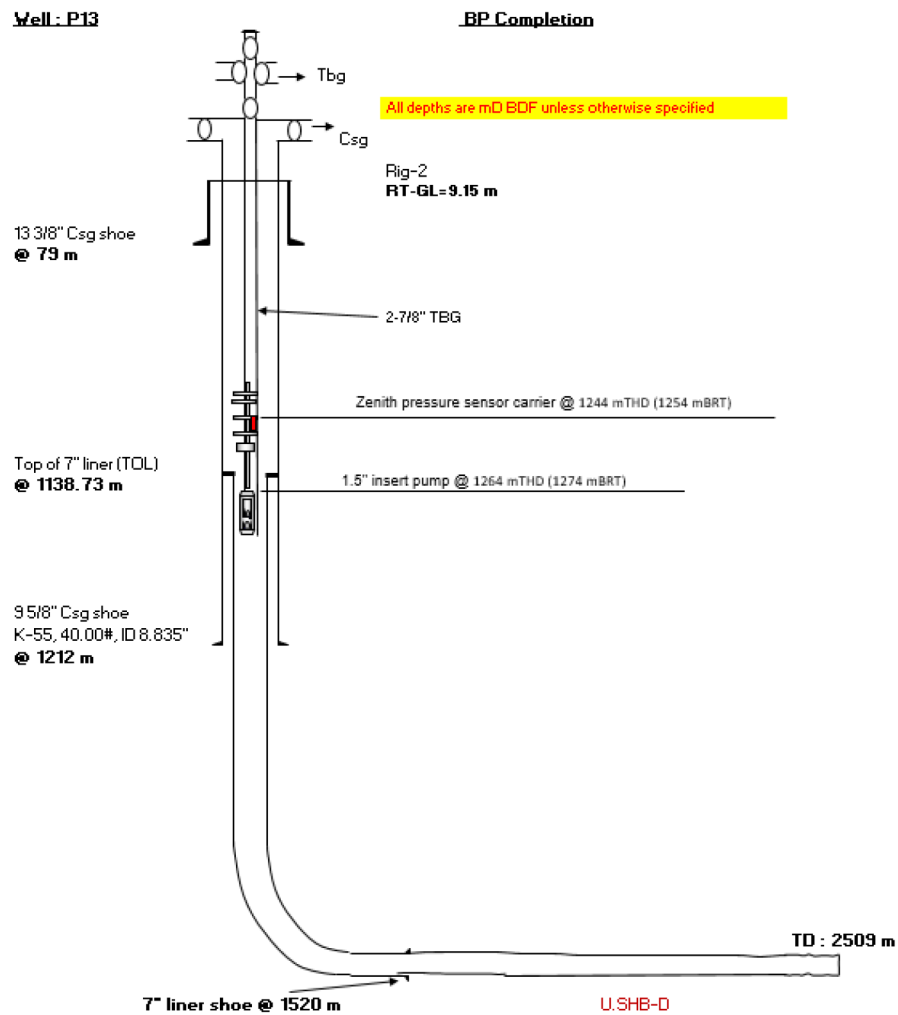


Figure 1—Example of BP Completion Diagram, Well P13

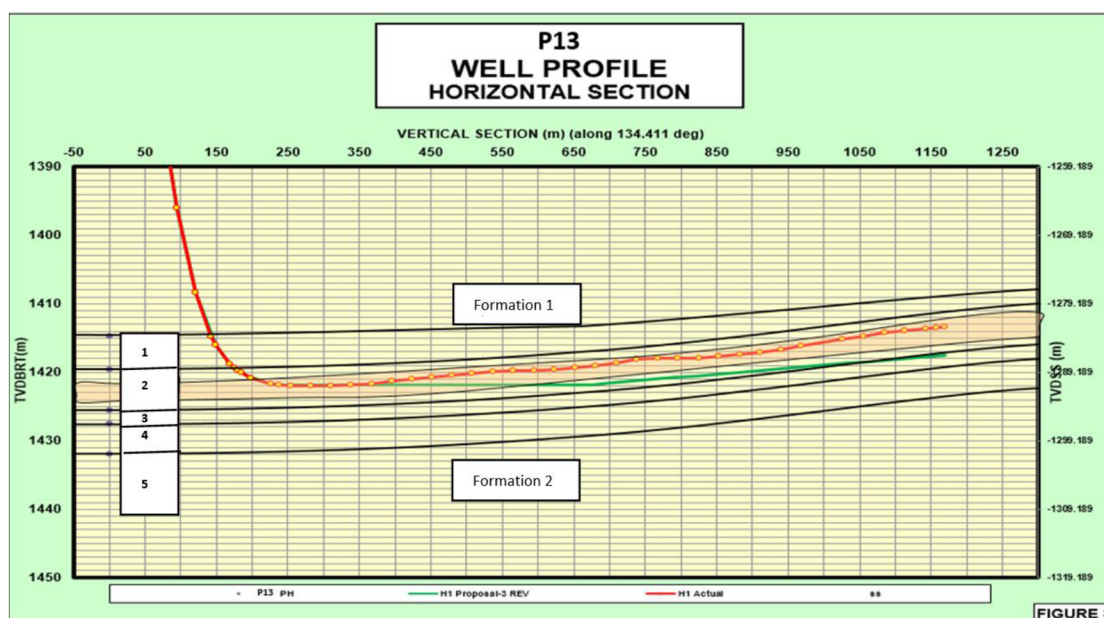


Figure 2—Example of Well Profile, well P13

Problem Description

The Field is a mature, high GOR carbonate reservoir with declining reservoir pressure, where artificial lift performance is increasingly challenged by gas interference. In many wells, large displacement sucker rod pumps currently in operation experience gas locks, fluid pound, and low pump fillage (Average PF is 51% and reduced volumetric efficiency of around 24%) These effects lead to unstable production rates, increased pump-off conditions, and higher rates of mechanical failure such as rod buckling and accelerated plunger/barrel wear, tubing wear.

Serious damage could be resulted due to Fluid Pound and Gas Interference

In case Fluid Pound or gas interference allowed to exist, extreme damage can occur to the entire system that can be the primary cause of the following equipment failures:

1. Fatigue failure of gear teeth and bearings.
2. Fatigue failure within the rod string.
3. Accelerated damage to the traveling valve and its cage. Valve rod breakage, barrel rupture, and standing valve failure can also occur.
4. "Fluid Pound" action accelerates wear of the tubing threads, causing leakage. It is frequently the cause of fatigue parting of the tubing.

Field diagnostics, including dynamometer analysis and PIP/fluid level surveys, have consistently shown that in wells with high GOR and low bottomhole pressure, larger pumps struggle to maintain consistent fillage. The mismatch between reservoir inflow capacity and pump outflow rate results in excessive free gas intake, unstable fluid columns, and significant downtime due to unscheduled workovers.

Reservoir Inflow vs. Pump Outflow Scenarios

In artificial lift operations, particularly with oversized pumps, the mismatch between reservoir inflow capacity and pump outflow rate can lead to significant operational challenges. The reservoir inflow, governed by the inflow performance relationship (IPR), is pressure-dependent and represents the maximum sustainable fluid delivery rate at given bottomhole pressures. Oversized pumps, designed to lift more than the reservoir can naturally supply, often induce excessive drawdown, pushing bottomhole pressure below

optimal levels. This can result in gas breakthrough, pump cavitation, gas locking (especially in ESPs and beam pumps). Graphical analysis through nodal plotting of IPR and pump performance curves clearly illustrates how increasing pump capacity shifts the operating point to lower pressures and may lead to unstable or inefficient production regimes. Proper sizing and dynamic monitoring of pump performance are therefore critical to maintaining reservoir health and optimizing production. Figure 3 shows how improving reservoir inflow or reducing outflow restrictions shifts IPR/VFP curves to unlock higher well productivity.

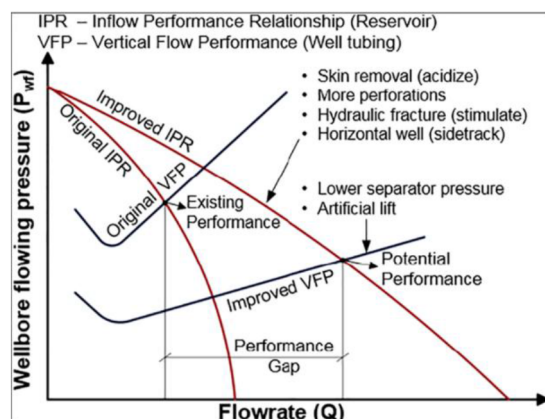


Figure 3—Inflow Performance Relationship Vs Vertical Flow Performance Curve

These issues not only reduce gross production but also increase operating costs through higher intervention frequency and equipment replacement. The challenge is to identify an artificial lift configuration—particularly in terms of pump sizing—that can effectively handle gas interference, maximize pump fillage, and extend run life without compromising production sustainability.

Normal Pumping Cycle

Figure 4 illustrates the sucker rod pump cycle. On the left, the ideal dynacard shows load variations on the polished rod during upstroke and downstroke, with valves opening and closing as rods stretch or compress. On the right, the pump operation is broken into stages—expansion, intake, compression, and discharge—demonstrating how fluid enters and is lifted during each stroke. This represents the normal pumping cycle without gas interference.

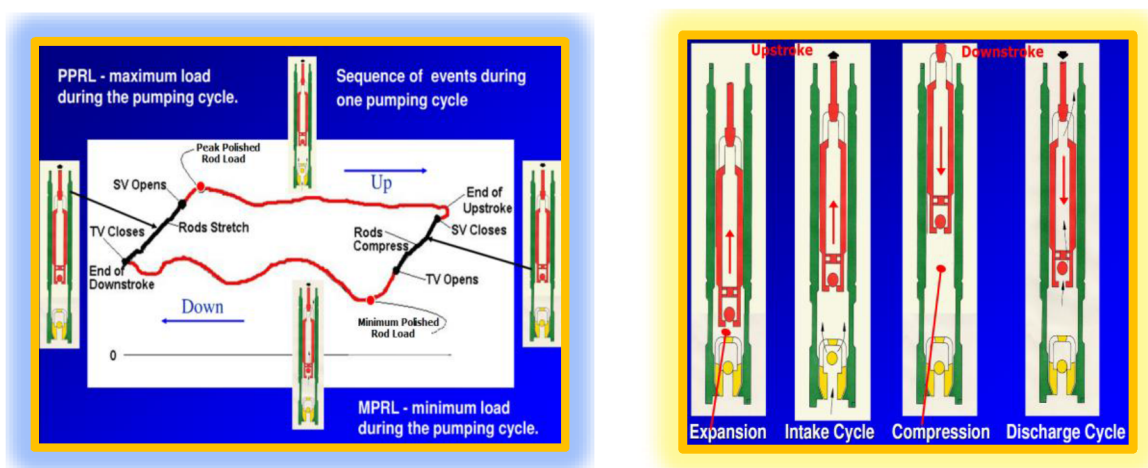


Figure 4—(a) Schematic Diagram of Ideal Dynacard, (b) Schematic Diagram of Pump Operation

Pumping Cycle with Gas Interference

Figure 5 illustrates the gas interference mechanism in sucker rod pumps. Gas enters the pump barrel and compresses during the downstroke, delaying the traveling valve opening until higher pressure is reached. This leads to partial fluid fillage, reduced effective load transfer, and the characteristic "gas pound" dynacard response.

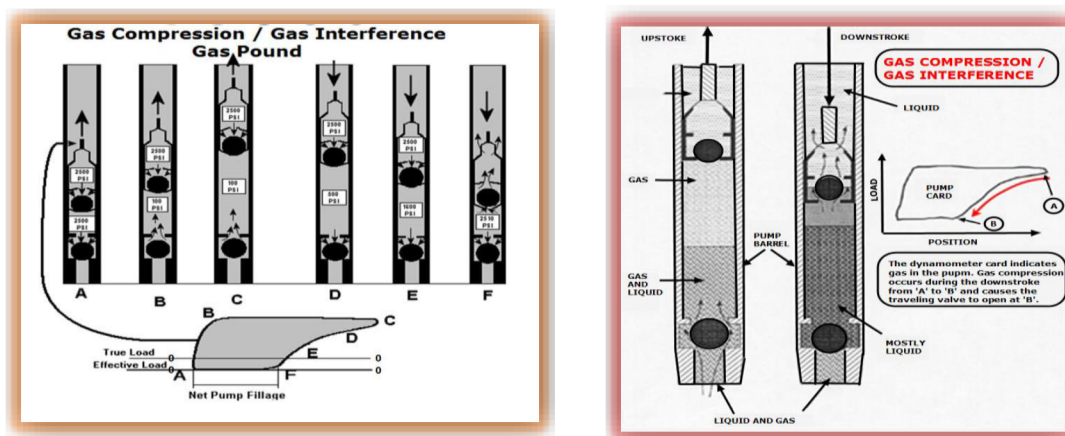


Figure 5—Gas Interference Mechanism, Echometer Company and Weatherford

Methodology

The design strategy was based on the hypothesis that using a smaller rod pump which is 1.25" & 1.5" pump with high clearance (0.01") in high GOR wells would enhance gas handling efficiency and pump reliability under gas interference conditions. The rationale included:

- **Improved Pump Fillage:** Smaller pumps are more capable of achieving higher fillage under gas interference conditions due to lower intake volume requirements per stroke. Figure 6 shows that smaller pumps improve liquid fillage and gas handling in high GOR wells compared to oversized pumps.
- **Improved Fluid Level Control:** Smaller pumps remove less volume per stroke, allowing the annular fluid level to stabilize above the pump intake, minimizing the influx of free gas into the pump.
- **Enhanced Compression Ratio:** By reducing the pump bore size, the compression ratio improves, increasing the ability of the pump to handle gas-laden fluids. Figure 7 shows that smaller pump sizes (e.g., 1-1/16" and 1-1/4" plungers) achieve higher compression ratios compared to larger pumps at the same stroke length. As pump size increases, the compression ratio decreases, making smaller pumps more effective in handling gas and improving pump efficiency in high GOR conditions.
- **Lower Risk of Mechanical Failure:** Higher pump fillage improves internal lubrication, reducing wear on the plunger and barrel. This minimizes friction, lowers the chance of rod buckling, and extends the life of downhole components.
- **Improved Volumetric Efficiency:** Higher fillage and reduced gas interference contribute to better volumetric efficiency, allowing more of the pump's displacement to translate into actual fluid production.

$$\eta_v = \frac{\text{Actual Pumped Volume}}{\text{Theoretical Pump Displacement Volume}} \times 100\%$$

- **Better Inflow–Outflow Matching:** In low-inflow wells, smaller pumps better match the reservoir inflow capacity with the pump outflow rate, minimizing pump-off events and improving system balance.
- **High clearance (0.01"):** it refers to the increased gap between the plunger and the pump barrel. This larger clearance provides a pathway for gas to escape more easily around the plunger during each pump cycle. By allowing gas to bypass the plunger through this gap, it reduces the buildup and compression of gas inside the pump, which minimizes gas lock. As a result, gas handling efficiency is significantly improved, leading to enhanced pump reliability and more consistent operation in high GOR wells.

Effect of Pump Size on Fillage in High GOR Wells

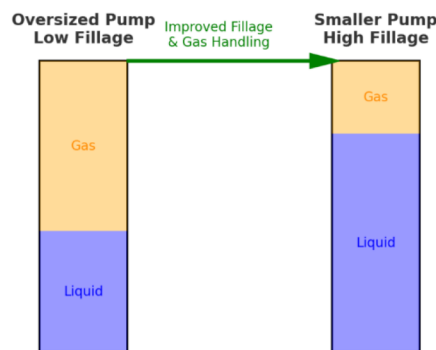


Figure 6—Effect of Pump size on Pump Fillage

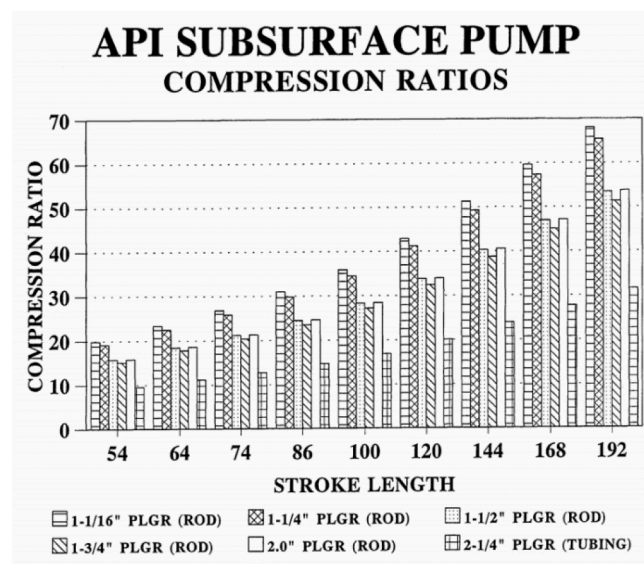


Figure 7—Downhole Pumps' Compression Ratios

The below specifications are for the pumps used in this project:

- 25-125-RAHM-24-4-2, 1.25" Insert Pump
- 25-150-RAHM-24-4-2, 1.50" Insert Pump

Well Selection Criteria

To ensure meaningful evaluation of smaller rod pumps with high clearance, candidate wells were carefully selected based on reservoir behavior, production characteristics, and operational challenges. Table 3 below outlines the key selection parameters and their relevance to identifying wells where smaller pumps are likely to offer improved performance:

Table 3—Smaller Pump Candidate Selection Criteria

Parameter	Recommended Range / Condition	Justification
API Gravity	30–42° API	Medium to light oil allows better pump-down control and gas handling.
Reservoir Pressure	BHP (< 300 psi)	Low pressure increases free gas entry; smaller pumps reduce drawdown rate.
Pump Fillage (Historical)	< 70–60% with existing oversized pump	Low fillage indicates mismatch between inflow and pump size.
Gas locking Frequency	Highly frequent	Direct indicator of poor gas separation and oversized pump issues
Fluid Production Rate	Low to moderate (< 150 bfpd)	Smaller pumps better match inflow in low-rate wells, reducing pump-off events.
Pump Efficiency	< 40% with oversized pump	Low downhole efficiency suggests gas-related inefficiencies.

Field Implementation

Following the selection process, a total of sixteen (16) wells were selected and equipped with smaller sucker rod pumps with high clearance (0.01"). These wells met the defined criteria, including high GOR, low pump fillage, and frequent gas interference. The new pumps were installed at nearly the same setting depths as their previous oversized counterparts to ensure consistency in drawdown and comparison of inflow performance.

While maintaining consistent pump depth, the operating parameters—specifically Stroke Per Minute (SPM) and Stroke Length (SL)—were not standardized across all wells. This was intentionally done to tailor the pump performance to each well's unique reservoir conditions and inflow capacity. Adjusting SPM and SL allows optimization of volumetric efficiency and avoids over-pumping, which could reintroduce gas locking or reduce runlife. This flexibility in operational tuning is essential to fully realize the benefit of smaller pump designs in variable high-GOR environments.

Data Collection and Analysis:

1. Pre- and post-installation production data were collected for each well over a monitoring period of 3–6 months.
2. Pump fillage, PIP, gross and production, and pump efficiency were evaluated.
3. Incidents of gas locking and pump-off controller behavior were compared.
4. Statistical analysis was performed to quantify improvements.

Results & Analysis

Tables 4, 5 & 6 are showing comparison of field performance parameters for oversized and smaller sucker rod pumps:

Performance Metrics for Oversized Pump Installations

Table 4—Oversized Pump Performance in in the Studied Field

Well	Runlife (days)	Average production rate (bbls/day)	Commissioning Date
P1	1408	70.8	07/02/2019
P2	284	89.1	04/03/2023
P3	745	21.3	18/11/2020
P4	1463	8.0	01/10/2020
P5	1774	69.8	30/11/2017
P6	871	7.1	28/11/2020
P7	333	10.6	15/05/2022
P8	358	5.1	15/04/2022
P9	643	90.3	12/09/2022
P10	721	14.5	07/12/2020
P11	858	48.8	09/07/2020
P12	737	68.6	08/08/2022
P13	2040	9.5	28/12/2018
P14	1379	31.9	06/01/2019
P15	911	79.4	31/03/2021
P16	1142	8.7	11/11/2021

Performance Metrics for small Pump Installations

Table 5—Smaller Pump Performance in the Studied Field

well	Commissioning Date	Runlife(days)	Average production rate (bbls/day)	Status
P1	12/19/2022	215	68.1	Elective Well
P2	9/26/2024	164	93.2	Failed
P3	1/29/2023	245	91.5	Failed
P4	10/9/2024	234	59.6	Running
P5	10/12/2022	962	115.5	Running
P6	5/5/2023	757	72.9	Running
P7	5/7/2023	755	63.7	Running
P8	5/5/2023	757	77.7	Running
P9	6/20/2024	345	66.5	Running
P10	12/1/2022	912	105.9	Running
P11	9/26/2023	613	54.0	Running
P12	10/14/2024	229	77.6	Running
P13	1/25/2025	126	71.0	Running
P14	10/31/2022	943	51.8	Running
P15	11/7/2023	571	132.2	Running
P16	12/31/2024	151	14.8	Running

Comparison of Performance Parameters for Oversized and Smaller Pumps

Table 6—Comparison of Oversized and Smaller Pumps

Well	Increase In Pump Fillage After Installing Smaller Pump %	Production Rate Difference	Production Rate Difference %	Comment
P1	N/A**	-2.7	-4%	-
P2	138%	4.2	5%	-
P3	-12	70.2	330%	-
P4	9%	51.5	643%	Well stimulation was performed with pump replacement (Acid Wash)
P5	27%	45.7	65%	-
P6	3%	65.8	929%	Well stimulation was performed with pump replacement (LVA*)
P7	20%	53.1	501%	Well stimulation was performed with pump replacement (Matrix Stimulation)
P8	29%	72.6	1432%	Well stimulation was performed with pump replacement (LVA*)
P9	26%	-23.9	-26%	-
P10	5%	91.4	632%	-
P11	7%	5.1	11%	-
P12	N/A**	9.0	13%	Well stimulation was performed with pump replacement (Acid Wash)
P13	9%	61.5	646%	Well stimulation was performed with pump replacement (Matrix Stimulation)
P14	21%	20.0	63%	-
P15	5%	52.9	67%	-
P16	42%	6.1	70%	-

LVA* : Large Volume Acidizing

N/A**: Data Unavailable

Production Rate Comparison between Oversized & Smaller Pumps

Figure 8 compares production rates between oversized and smaller pumps across multiple wells. In most cases, smaller pumps deliver higher or more stable production rates than oversized pumps, confirming their advantage in optimizing well performance, especially in high GOR conditions.

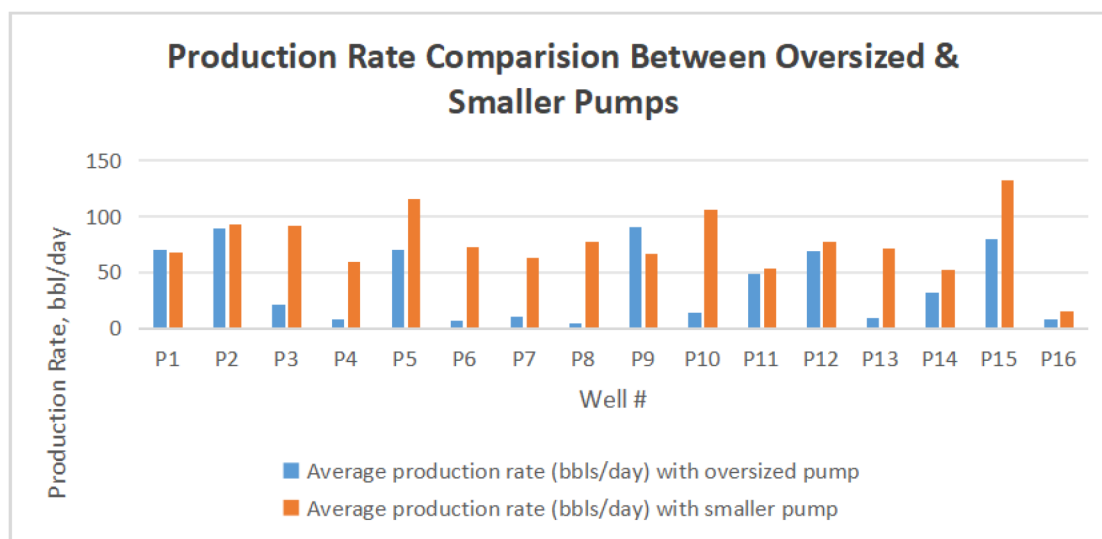


Figure 8—Production Rate Comparison between Oversized & Smaller Pumps in the Study

Volumetric Efficiency of The Large Vs Smaller Pumps in The Same Wells

Figure 9 compares volumetric efficiency (PE%) of large and smaller pumps in in wells from the field under study. Results show that smaller pumps consistently achieve much higher PE% than larger pumps, confirming their effectiveness in improving pump fillage and overall efficiency.

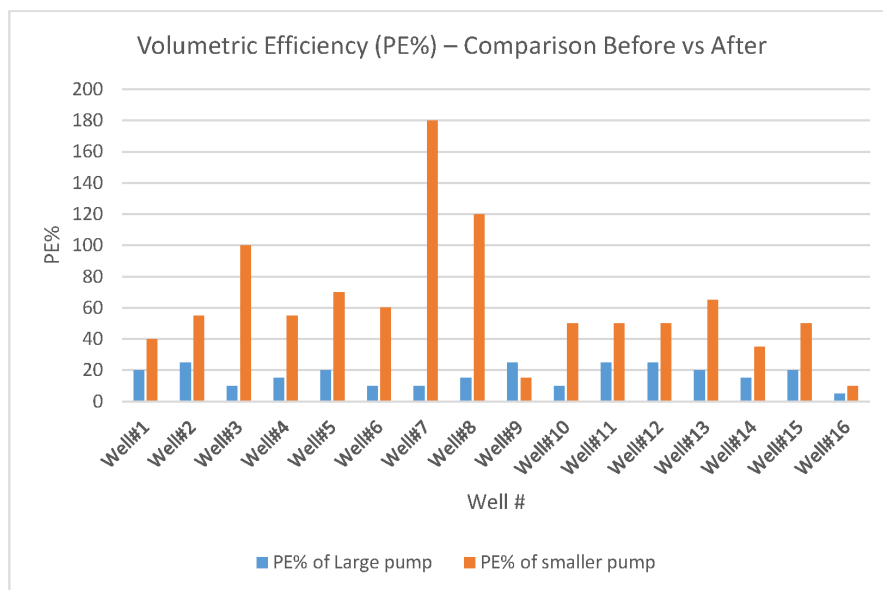


Figure 9—Volumetric Efficiency of the Large & Smaller Pumps in the Studied Field

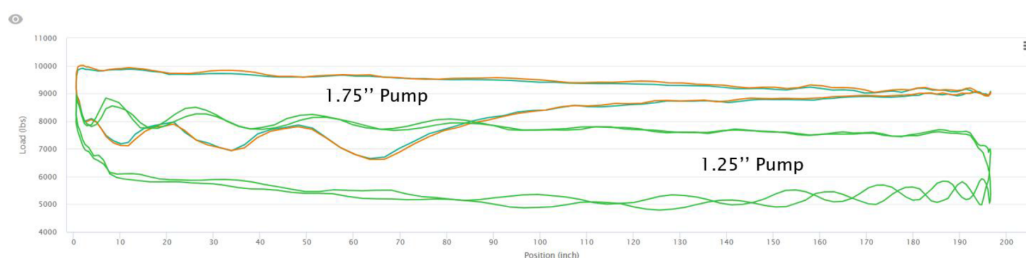
Statistical Comparison between Larger and Smaller Pump Performance for Three Wells

The three wells presented in table 7 were selected randomly solely to illustrate detailed performance parameters for wells that demonstrated notable improvements. These examples highlight key metrics such as pump fillage, downhole pump efficiency, and production rate before and after switching from larger to smaller pumps. The selection aims to provide a clear comparison of the benefits observed in wells with good performance gains, without implying any specific pattern or preference in the choice of wells.

Table 7—Statistical Comparison (Well P5, 14 and 15)

Well P5		
Metric	Larger Pump (Before)	Smaller Pump (After)
Pump Fillage (%)	48.1	61
Downhole Pump Efficiency	33	74
Production Rate (BPD)	69	115
Well P14		
Metric	Larger Pump (Before)	Smaller Pump (After)
Pump Fillage (%)	68.3	82.6
Downhole Pump Efficiency	15	32
Production Rate (BPD)	31.9	51.8
Well P15		
Metric	Larger Pump (Before)	Smaller Pump (After)
Pump Fillage (%)	83	88
Downhole Pump Efficiency	37	85
Production Rate (BPD)	79.4	132.2

Example 1



Example 2

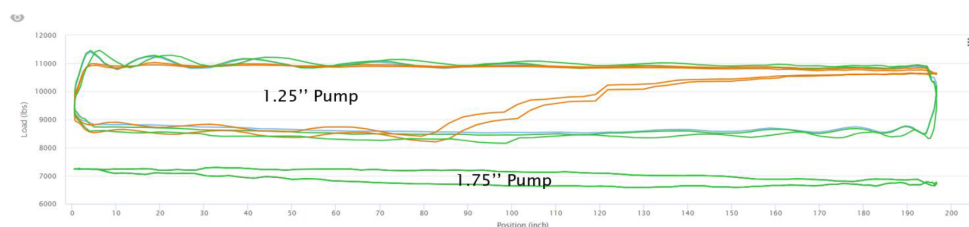


Figure 10—Comparative Pump Dynacards of (Well P5 and 15)

Well#1 Dynacards

Before with 1.75\" Pump: Dynacards shows gas interference with less fluid load.

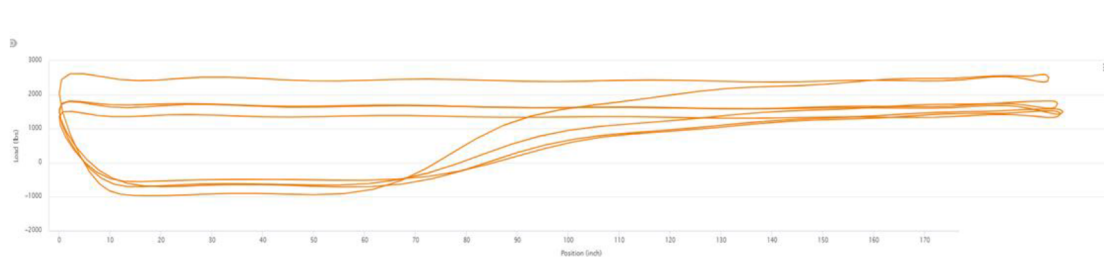


Figure 11—Dynacard of Well # 1 (Large Pump)

After with 1.25\" Pump: Improved Pump fillage compared with previous 1.75\" pump cards, and also fluid load improved.

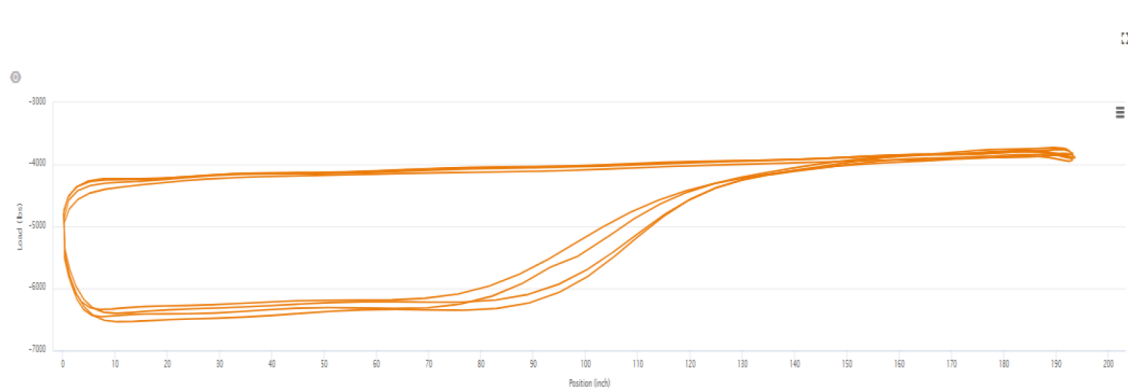


Figure 12—Dynacard of Well # 1 (Small Pump)

Well#2 Dynacards

Before with 1.75" Pump: Dynacards shows gas interference with less fluid load.

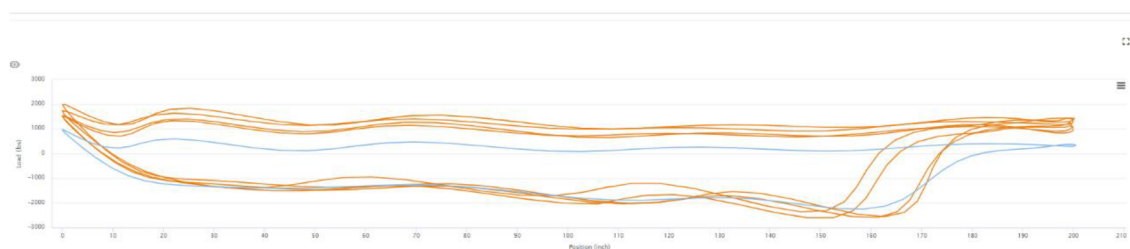


Figure 13—Dynacard of Well # 2 (Large Pump)

After with 1.50" Pump: Improved Pump fillage compared with previous 1.75" pump cards, and also fluid load improved.

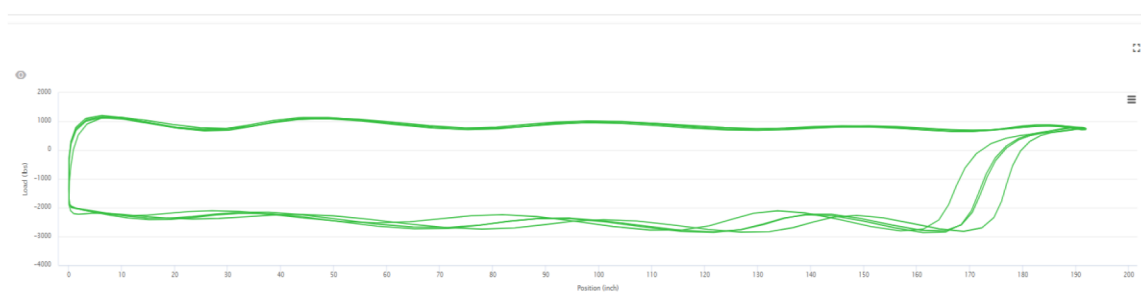


Figure 14—Dynacard of Well # 2 (Small Pump)

Discussion

Interpretation

The comparison between oversized and smaller sucker rod pumps in operator's high GOR wells shows encouraging performance trends, but runlife results for most wells are still incomplete, as the majority are in "Running" status. This means that current runlife figures for these wells represent partial durations and will continue to increase until the failure. Therefore, final conclusions on runlife improvement or reduction can only be drawn for wells where the smaller pump has already failed or was pulled for elective reasons (e.g., P1, P2, P3).

Production rate improvements, however, can be evaluated immediately. Many wells equipped with smaller pumps show significant production gains, particularly in low-inflow, high-GOR conditions where oversized pumps suffered from gas interference (e.g., P6, P7, P8, P10). These gains suggest that smaller pumps are better matched to the reservoir inflow capacity, maintain higher pump fillage, and reduce gas locking.

Table 6 is showing that out of the 16 wells studied, 14 wells showed improvement in oil gain after downsizing to smaller pumps. Among these, 8 wells had exactly similar parameters and pump setting depths, making pump size the only variable influencing the results. However, in 6 of the 14 wells, acid stimulation treatments (acid wash, LVA, or matrix stimulation) were conducted alongside the pump replacement. These stimulations are believed to have contributed to the additional oil gain, in combination with the impact of installing smaller pumps.

Some wells (e.g., P1, P9) recorded lower production rates after downsizing, which could be due to lack of pressure support.

Overall, early results point to smaller pumps offering potential for increased production stability and better gas handling in suitable candidate wells, with the expectation that ongoing wells will provide more conclusive runlife data over time.

The analysis demonstrates that downsizing to smaller pumps significantly improved well performance across multiple aspects. By reducing displacement volume, pump fillage increased even in low inflow wells, minimizing fluid pound and rod string stresses. Higher compression ratios in smaller pumps also enhanced gas handling by limiting gas accumulation, slippage, and compression effects, thereby lowering the risk of gas locking. The improved inflow–outflow balance ensured the pump intake remained submerged, preventing gas breakthrough and stabilizing production in low PI wells. Additionally, smaller pumps reduced mechanical stresses such as peak rod loads, torque, and power demand, which directly translates into longer runlife and lower maintenance. Finally, the gentler drawdown strategy optimized reservoir drainage by mitigating early gas coning and maintaining more favorable bottom-hole pressures, ultimately supporting more stable and efficient production.

Table 8—Summary of the Studied Pumps Performance

Factor	Larger Pump (1.75", 2.25" & 2.75")	Smaller Pump (1.25" & 1.50")
Volumetric Efficiency	Low (<40%)	High (≥52%)
Gas Lock Risk	High	Low
Stroke Efficiency	Poor	Improved
Rod Load / Torque	High	Reduced
Fluid Level Stability	Poor (drawdown too fast)	Better (draws at matched rate)

Field Learnings

- **Improved Gas Handling:** Smaller displacement pumps with high clearance consistently maintained higher pump fillage under high GOR conditions, reducing gas locking incidents.
- **Enhanced Volumetric Efficiency:** Better match between pump capacity and reservoir inflow improved volumetric efficiency and stabilized production rates.
- **Operational Stability:** Smaller pumps maintained steady performance even at declining reservoir pressure, providing predictable production profiles.
- **Design Optimization Insight:** Highlighted the importance of matching pump size not only to production potential but also to gas interference characteristics for long-term sustainability.

Advantages of Installing Smaller Pumps in Horizontal Sections

Beyond their use in the vertical section, smaller pumps have also been deployed in the tangent (horizontal) section, where they demonstrated the following advantages:

- **Better Fit for Curved Geometry:** Smaller pumps are easier to deploy and align within the tight bends and curvature of horizontal wellbores.
- **Reduced Installation Risk:** Compact size minimizes the chance of mechanical interference or equipment damage during installation and retrieval.
- **Minimized Rod Buckling:** Lighter rod strings experience less compression and buckling in the deviated or horizontal section.
- **Improved Efficiency in Low-Inflow Reservoirs:** Smaller pumps match low production rates better, leading to higher pump fillage and more consistent fluid lifting.

- **Delayed Gas Interference:** Smaller chambers can help delay free gas entry, maintaining more stable liquid production.

Conclusion

The results from this field study demonstrate that smaller-capacity sucker rod pumps with increased plunger clearance offer significant advantages in high GOR, low BHP environments commonly found in the operator's mature oil field. The downsized pumps consistently improved pump fillage, enhanced volumetric efficiency, and reduced gas locking incidents. This led to more stable production performance and, in many cases, extended equipment runlife.

Although runlife data remain ongoing for most wells, early indicators show that the optimized inflow-outflow matching achieved by smaller pumps can defer mechanical failures and reduce operational costs. The ability to maintain higher dynamic fluid levels above the pump intake while minimizing drawdown rate was crucial in controlling gas entry and improving overall system performance.

This study validates the importance of matching pump displacement to reservoir inflow and reinforces the role of smaller pumps as a viable optimization strategy for artificial lift systems in gas-prone, low-rate wells.

Recommendations

Screen candidate wells exhibiting the following issues:

- Low pump fillage
- Frequent gas locking
- Declining dynamic fluid levels

Incorporate dynamic pump sizing into routine WRFM workflows and artificial lift optimization plans, especially in high GOR or low inflow environments.

Further Work

- **Extended Monitoring Period:** To better assess the long-term performance, a longer observation period is required to accurately evaluate runlife trends and detect any delayed failures.
- **Failure Analysis and Classification:** Investigate the nature of pump failures (e.g., mechanical wear, gas locking, plunger damage) and determine if these are specifically related to pump downsizing or are independent operational issues.
- **Impact on Rod and Tubing Integrity:** Monitor potential effects of smaller pump operations on rod buckling, tubing wear, and overall mechanical integrity over time.
- **Economic Assessment:** Perform a full cost-benefit analysis that includes workover cost savings, production uplift, and runlife extension to evaluate overall project value.

Abbreviations and Glossary

Abbreviation	Full Form	Description
API	American Petroleum Institute	Refers to API gravity (oil density measure) and industry standards.
BHP	Bottom-Hole Pressure	Fluid pressure measured at pump setting depth.
BPD (bbl/day)	Barrels per Day	Standard unit of oil production rate.
ESP	Electrical Submersible Pump	Alternative artificial lift method to SRP.

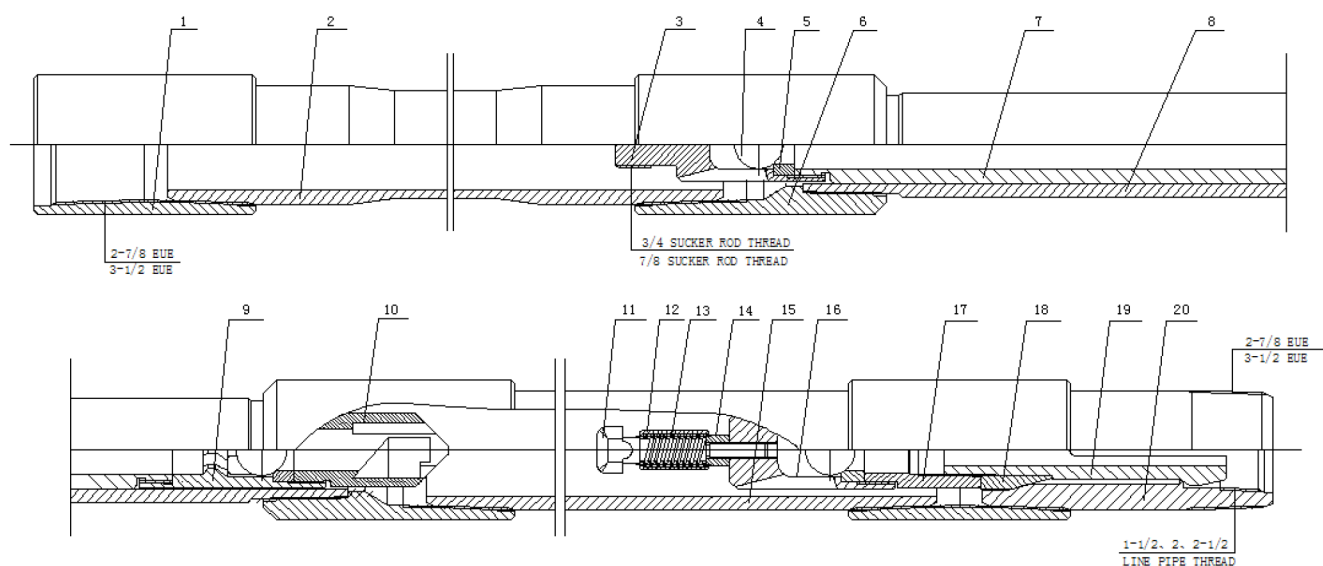
Abbreviation	Full Form	Description
GOR	Gas–Oil Ratio	Volume of gas produced per barrel of oil (scf/STB).
IPR	Inflow Performance Relationship	Reservoir deliverability curve versus bottom-hole pressure.
OPEX	Operating Expenditure	Ongoing cost of field production operations.
PF	Pump Fillage	% of pump chamber filled with liquid per stroke.
PE	Pump Efficiency / Volumetric Efficiency	Ratio of actual liquid lifted to theoretical pump displacement.
PIP	Pump Intake Pressure	Pressure of fluids entering the pump chamber.
RPM	Revolutions per Minute	Rotational speed of the pumping unit motor.
SL	Stroke Length	Distance traveled by the plunger per stroke.
SPM	Strokes per Minute	Pumping speed, number of strokes per minute.
SRP	Sucker Rod Pump	Beam pumping artificial lift system.
TVD	True Vertical Depth	Vertical depth from surface to reservoir or pump.
VFP	Vertical Flow Performance	Pressure–flow performance relationship in tubing.
WRFM	Well, Reservoir, and Facility Management	Integrated optimization workflow for production.

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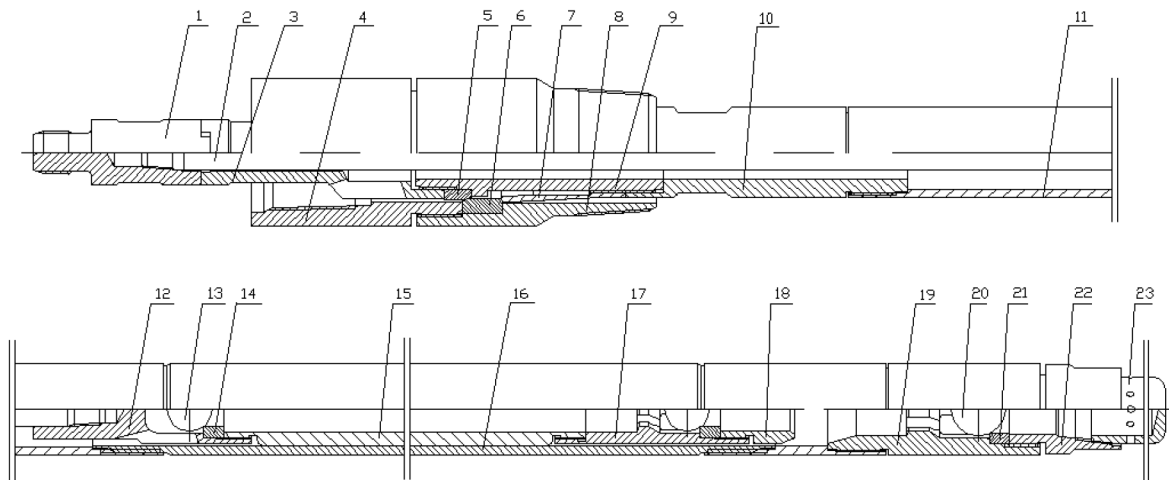
Appendix

Technical Structural Drawing for THM Pump



Item	Specification	Material	Quantity
1	Coupling, Crossover	45# (AISI 1045)	2
2	Nipple, Extension, Upper	45# (AISI 1045)	2
3	Cage, Top Open	40Cr with stellite alloys Line	1
4	Coupling, Barrel	45#(AISI 1045)	2
5	Valve Ball Φ28.58	Titanium Carbide	3
6	Valve Seat 175	Tungsten carbide	3
7	Plunger, One Piece Pin (spraying metal)	45# (AISI 1045)	1
8	Barrel, Chrome Plated	20# (AISI 1020)	1
9	Cage, Closed Plunger Pin	40Cr with stellite alloys Line	1
10	Puller, Plunger Assembly	40Cr	1
11	Pin, Standing Valve Assembly	40Cr	1
12	Cap, Standing Valve Assembly	40Cr	1
13	Spring, Standing Valve Assembly	Monel	1
14	Locking Nut, Standing Valve Assembly	40Cr	1
15	Nipple, Extension, Lower	45#(AISI 1045)	1
16	Cage, Standing Valve	40Cr with stellite alloys Line	1
17	Bushing, Mechanical Bottom Lock	40Cr	1
18	Ring, Mechanical Bottom Lock	brass	1
19	Mandrel, Mechanical Bottom Lock	65Mn/60Si2Mn	1
20	Nipple, Seating, Mechanical Bottom	40Cr	1

Technical Structural Drawing For RHAM Pump



Item	Specification	Material	Quantity
1	Bushing, Valve Rod	40Cr	1
2	Rod, Valve	7/8" Sucker Rod	1
3	Guide, Valve Rod	40Cr	1
4	Nipple, Upper Half, Mechanical Top Lock	45# (AISI 1045)	1
5	Ring, Mechanical Top Lock	3Cr13	1
6	Seating, Mechanical Top Lock	3Cr13	1
7	Sleeve, Mechanical Top Lock	65 Mn/ 60Si2Mn	1
8	Nipple, Lower Half, Mechanical Top Lock	45# (AISI 1045)	1
9	Mandrel, Mechanical Top Lock	45# (AISI 1045)	1
10	Bushing, Mechanical Top Lock	40Cr	1
11	Coupling, Extension	45# (AISI 1045)	2
12	Cage, Top Plunger	40Cr with stellite alloys Line	1
13	Valve Ball, Traveling	Titanium Carbide	2
14	Valve Seat, Traveling	Tungsten carbide	2
15	Plunger, One Piece, Pin End	45# (AISI 1045)	1
16	Barrel, Heavy Wall	20# (AISI 1020)	1
17	Cage, Closed, Pin Plunger	40Cr with stellite alloys Line	1
18	Plug Seat	40Cr	1
19	Cage, Closed Barrel	40Cr with stellite alloys Line	1
20	Valve Ball, Standing	Titanium Carbide	1
21	Valve Seat, Standing	Tungsten carbide	1
22	Bushing, Seat, Barrel Cage	40Cr	1
23	Screen Pipe	40Cr	1